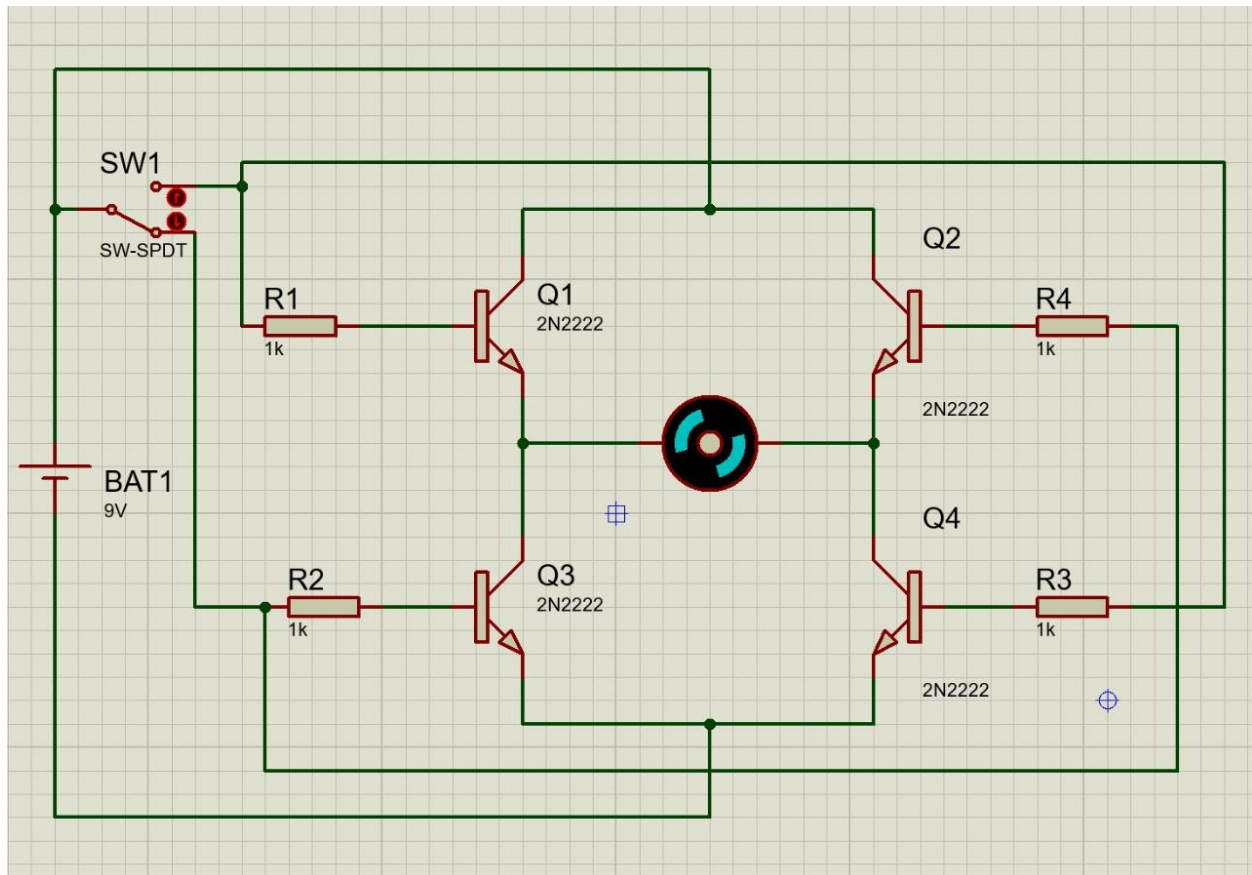


Autonomous

ID: 445236

Task 1:

Circuit on Proteus:



Components:



Explanation:

This is a BJT H Bridge for a dc motor that uses:

1. Battery

- 9 volt that supply the bridge and motor
- Has high internal resistance and low current best for small motors or low power need

2. SPDT

- Single Pole Double Throw
- Direct Control
- It has one input that can connect to one of two outputs and changes with a switch allowing the motor to rotate clockwise or counterclockwise once switched

3. Resistors

- Limit base current so base emitter not over driven

4. NPN Transistor

- It's a bipolar junction transistor with layers NPN that connect collector to emitter when base emitter is forward biased
- C: Collector, B: Base, E: Emitter
- The base current controls the collector current

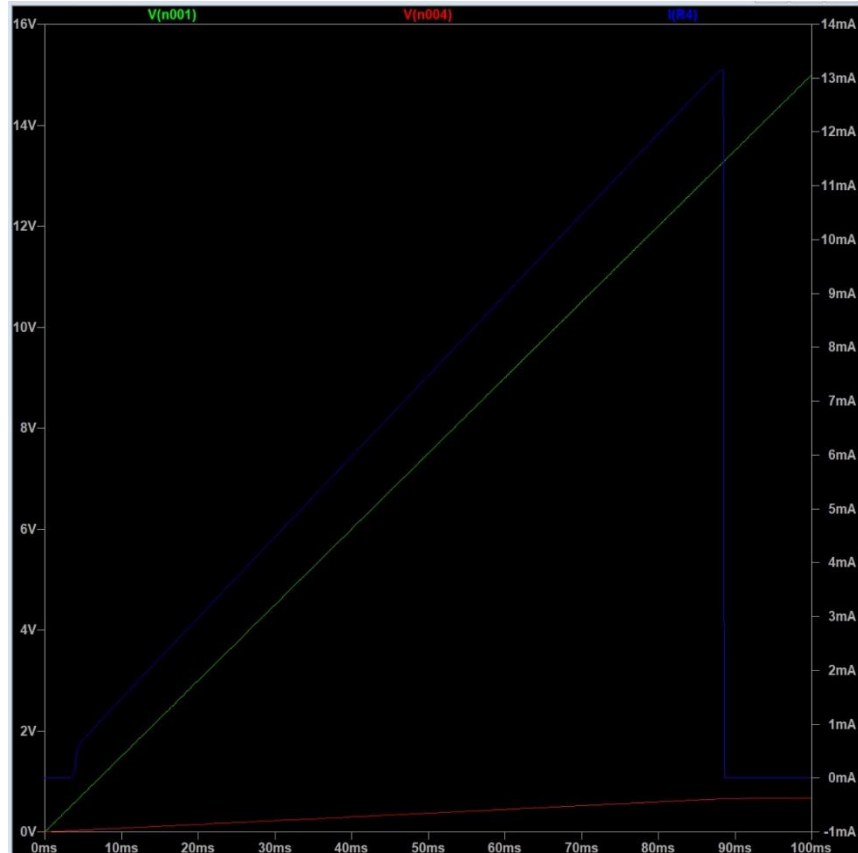
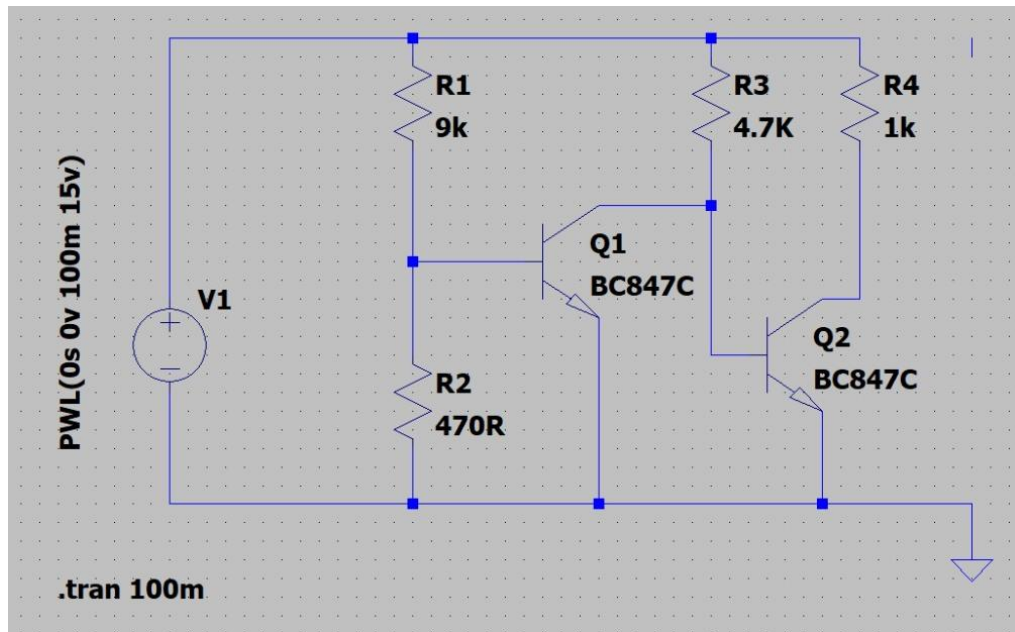
- At top are wired as emitter to collect 9v then send it to motor, their base rise and voltage in motor is $9 - 0.7 = 8.3\text{v}$
- At bottom are wired as emitters to ground so when base is high, they saturate and pull motor near ground
- The series connection between upper and lower BJT is a one-way current pass for motor to see the battery (9v) minus 1.4v
- Basically it only works diagonally at a time while current always go through ones connected in series
- MOSFETs are sometimes preferred cause of lower drop and cooler during run time

5. Motor

- Its an inductive load, where current is interrupted and emf is generated that uses motor as a path

Task 2:

NPN BJT Over Voltage Protection Circuit 1:



The load R4 is feed and supply ramp up then load cut off when supply exceeds a triple voltage, knowing that Q1 is series low side switch for load and Q2 is low side series switch for load. When V_{in} rise above certain point, Q1 turns on allowing base current to flow from Q2 and turning Q2 off dropping load current to zero.

Components:

1. Power Supply (V1):

- PWL 0 => (0 to 15v in 100ms)
- Input supply ramped to see when protection happens

2. Resistors (9k, 470, 4.7k, 1kohms):

- Load to adjust voltage
- R1 & R2 When base volt reaches base emitter volt, Q1 turns on
- R3 provide base current to Q2 turning it on when Q1 is off, its value perfect for not wasting current and saturation of Q2 at wanted load
- R4 monitor current that rise with V_{in} and when protection occurs is known when it goes to zero at once.

3. NPN Q1:

- Act like one transistor comparator
- Up flow Q1 sinks current from node to ground and Q2 is off
- Down flow Q1 is off so not to feed node base of Q2

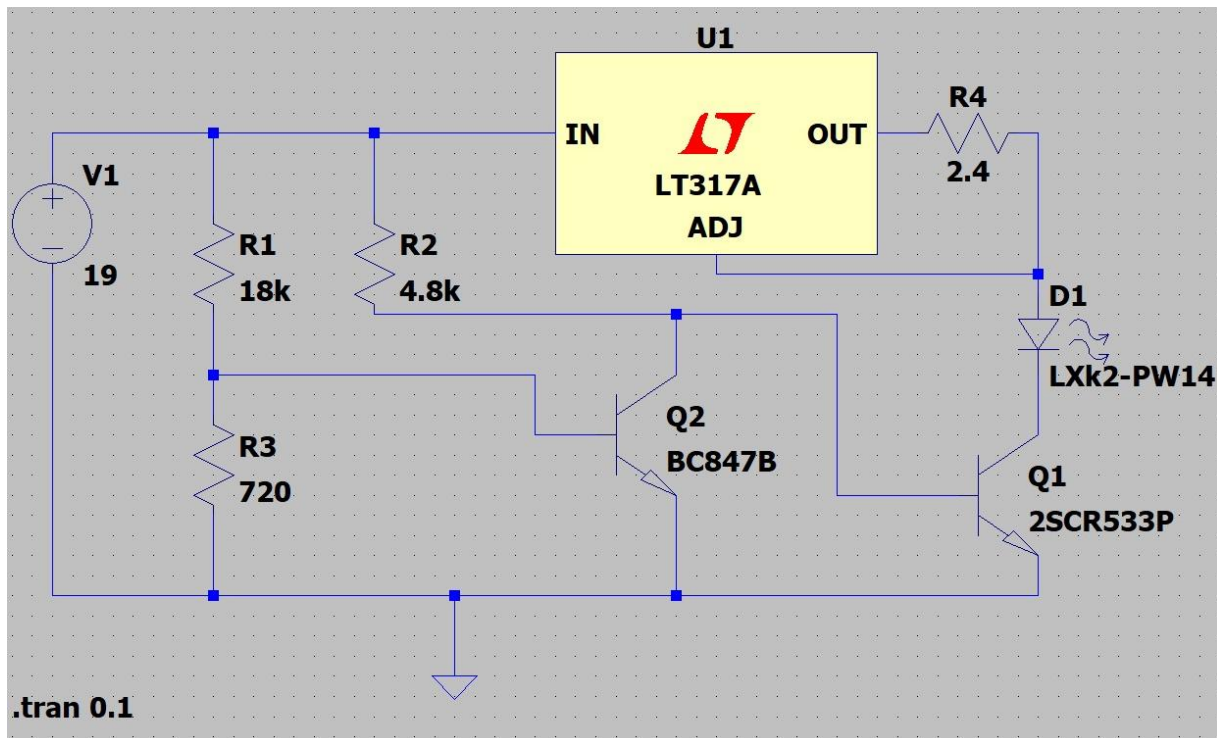
4. NPN Q2:

- Series switch that carry load current when turned on and with base current Q2 saturates and current load is $(V_{in} - V_{ce})/R4$ and Q1 is off

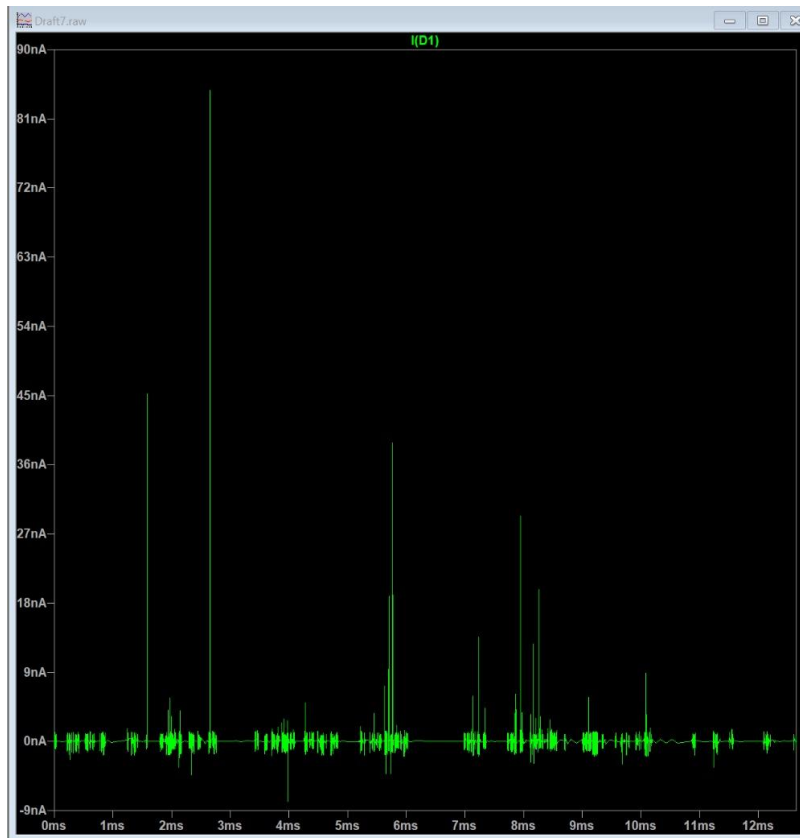
Graph:

- **Green** shows V_{in} that goes linearly up from 0v to 15v
- **Blue** shows load current increasing linearly then falls to 0mA when Q1 is on & Q2 is off
- **Red** shows node is low when Q1 tripped

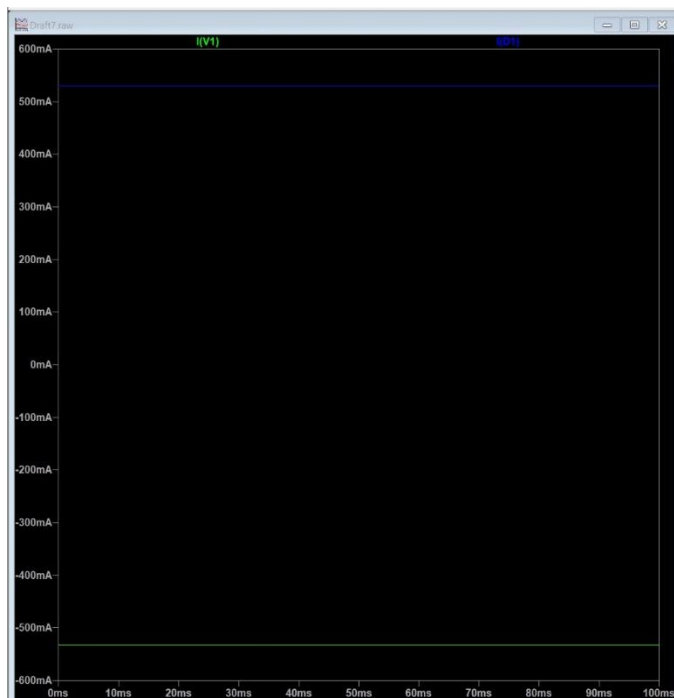
NPN BJT Over Voltage Protection Circuit 2:



When voltage was 19v

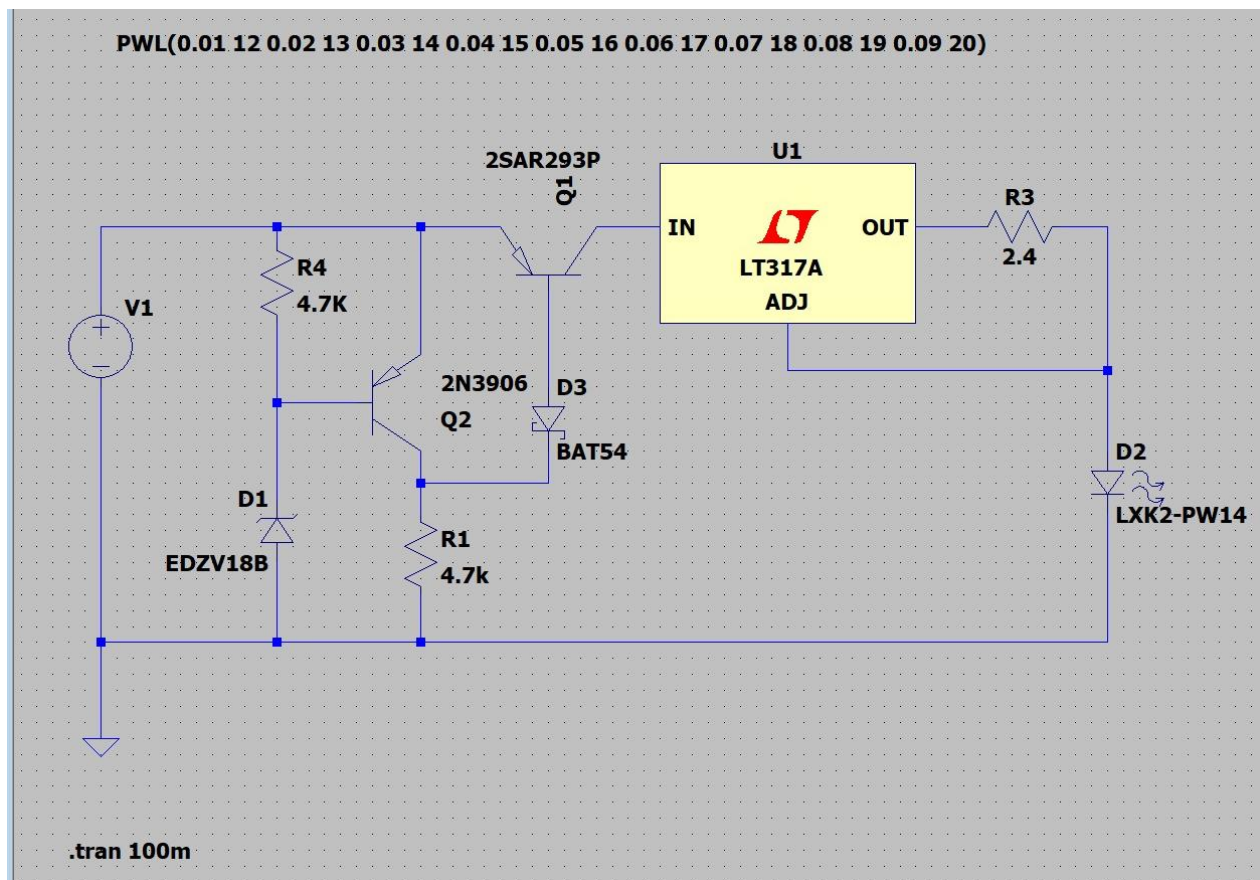


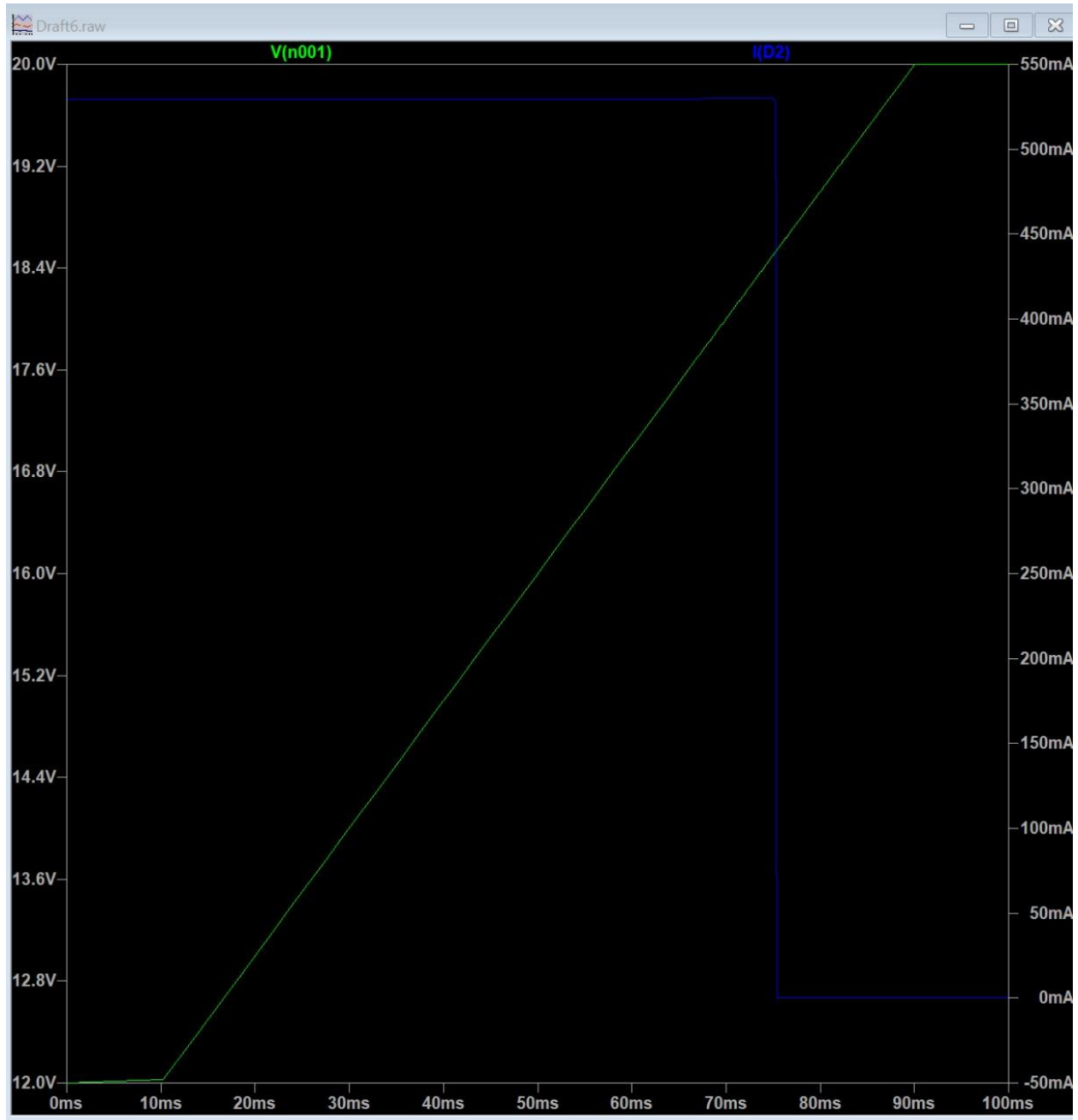
When voltage was 13v



This circuit is a constant-current LED driver with a high voltage safety. The LT317A and the small resistor beside it set a fixed LED current. Q1, NPN at bottom, is just switch that completes the LED's path to ground; R2 feeds Q1's base to stay on. The three resistors on the left watch the input voltage and drive the small transistor Q2. Knowing that the input is normal, Q2 is off, so Q1 stays on and the LED gets its regulated current. If the input rises too high, Q2 turns on and yanks the control node low, which turns Q1 off and opens the LED's return path. The LT317A still regulates current, but it's diverted internally instead of through the LED, protecting the LED from over-voltage. When the input drops back to normal, Q2 turns off again, Q1 turns back on, and the LED current resumes.

PNP BJT Over Voltage Protection Circuit:





it is a constant-current LED driver that turn off power if higher than normal. The LT317A with R3 sets the LED current. The PNP Q1 sits before the regulator; when Q1 is on, the driver gets power,

]and everything is normal. The little block on the left watches the supply: D1 is an 18 V Zener, R4 feeds it, R1 provides a return/bleed path, Q2 is a small PNP, and D3 is a one-way link to Q1's base.

- Normal voltage: the Zener is off → Q2 is off. Through R1 and D3, Q1's base is pulled a bit below its emitter, so Q1 turns on and the LED sees the regulated current.
- Over-voltage: the Zener conducts and drives Q2 on. Q2 pushes Q1's base up toward its emitter, removing its base-emitter bias, so Q1 turns off and cleanly disconnects power to the LT317/LED. The LED current drops to (near) zero, which is what your plot shows around ~19 V.
- When the supply falls back, the Zener and Q2 turn off, the base is pulled down again through R1/D3, Q1 turns back on, and the LED resumes normal current.

In short: Q1 is the high-side switch, D1-R4-R1-Q2-D3 are the voltage watcher, and together they cut power to protect the LED whenever the input climbs too high.